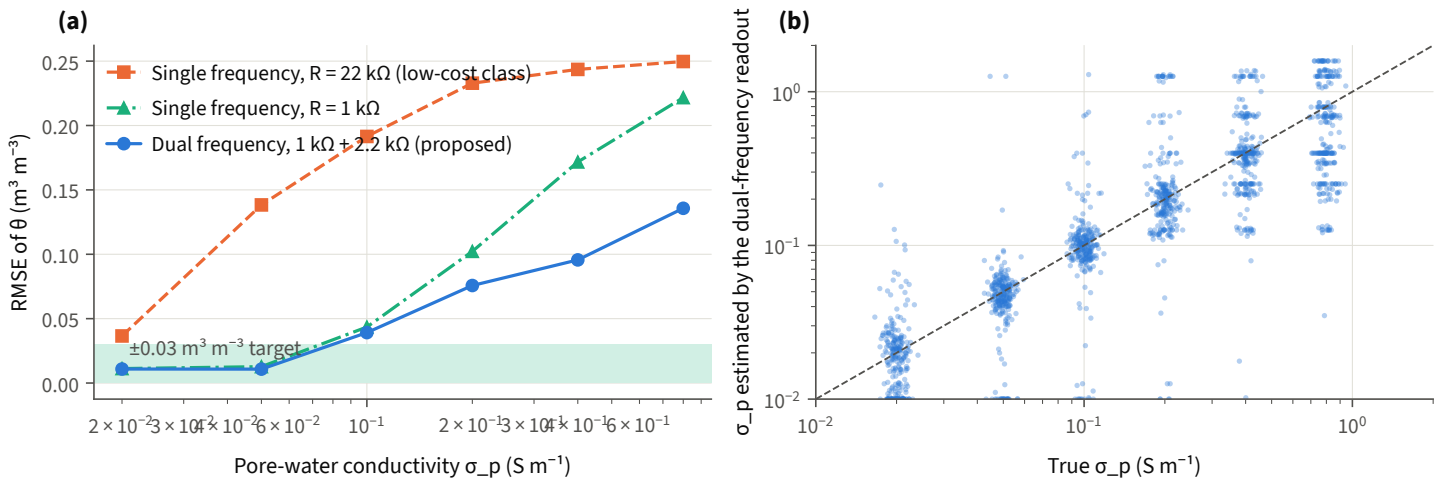
(a)
Single frequency, R = 22 kΩ (low-cost class)
Single frequency, R = 1 kΩ
Dual frequency, 1 kΩ + 2.2 kΩ (proposed)
±0.03 m³ m⁻³ target
RMSE of θ (m³ m⁻³)
0.25
0.20
0.15
0.10
0.05
0.00
2 × 10⁻²
3 × 10⁻²
4 × 10⁻²
6 × 10⁻²
10⁻¹
2 × 10⁻¹
3 × 10⁻¹
4 × 10⁻¹
6 × 10⁻¹
Pore-water conductivity σ_p (S m⁻¹)
(b)
σ_p estimated by the dual-frequency readout
10⁰
10⁻¹
10⁻²
10⁻²
10⁻¹
10⁰
True σ_p (S m⁻¹)